

```

> restart
> # orbits of the planar systems
> # non-linear because x^2, xy
> # Ex1
> # a) find all equilibrium points - set each derivative = 0
> solve({2*x-x*x-x*y},{-y+x*y});
Warning: solving for expressions other than names or functions is
not recommended.
> # b) jacobian matrix for each equilibrium
> #? Jacobian
> with(linalg)
[BlockDiagonal, GramSchmidt, JordanBlock, LUdecomp, QRdecomp,
Wronskian, addcol, addrow, adj, adjoint, angle, augment, backsub, band,
basis, bezout, blockmatrix, charmat, charpoly, cholesky, col, coldim,
colspace, colspan, companion, concat, cond, copyinto, crossprod, curl,
definite, delcols, delrows, det, diag, diverge, dotprod, eigenvals,
eigenvalues, eigenvectors, eigenvects, entermatrix, equal, exponential,
extend, ffgausselim, fibonacci, forwardsub, frobenius, gausselim,
gaussjord, geneqns, genmatrix, grad, hadamard, hermite, hessian, hilbert,
htranspose, ihermite, indexfunc, innerprod, intbasis, inverse, ismith,
issimilar, iszero, jacobian, jordan, kernel, laplacian, leastsqrs, linsolve,
matadd, matrix, minor, minpoly, mulcol, mulrow, multiply, norm, normalize,
nullspace, orthog, permanent, pivot, potential, randmatrix, randvector,
rank, ratform, row, rowdim, rowspace, rowspan, rref, scalarmul,
singularvals, smith, stackmatrix, submatrix, subvector, sumbasis, swapcol,
swaprow, sylvester, toeplitz, trace, transpose, vandermonde, vecpotent,
vectdim, vector, wronskian]
(1)

> with(DEtools)
[AreSimilar, Closure, DENormal, DEplot, DEplot3d, DEplot_polygon, DFactor,
DFactorLCLM, DFactorsols, Dchangevar, Desingularize, FindODE,
FunctionDecomposition, GCRD, Gosper, Heunsols, Homomorphisms,
IVPsol, IsHyperexponential, LCLM, MeijerGsols,
MultiplicativeDecomposition, ODEInvariants, PDEchangecoords,
PolynomialNormalForm, RationalCanonicalForm, ReduceHyperexp,
RiemannPsols, Xchange, Xcommutator, Xgauge, Zeilberger, abelsol,
adjoint, autonomous, bernoullisol, buildsol, buildsym, canoni, caseplot,
casesplit, checkrank, chinisol, clairautsol, constcoeffsols, convertAlg,
convertsys, dalembertsol, dcoeffs, de2diffop, dfieldplot, diff_table,
diffop2de, dperiodic_sols, dpolyform, dsubs, eigenring,

```

(2)

endomorphism\_charpoly, equinv, eta\_k, eulersols, exactsol, expsols, exterior\_power, firint, firtest, formal\_sol, gen\_exp, generate\_ic, genhomosol, gensys, hamilton\_eqs, hypergeometricsols, hypergeomsols, hyperode, indicialeq, infgen, initialdata, integrate\_sols, intfactor, invariants, kovacicsols, leftdivision, liesol, line\_int, linearsol, matrixDE, matrix\_riccati, maxdimsystems, moser\_reduce, muchange, mult, mutest, newton\_polygon, normalG2, ode\_int\_y, ode\_y1, odeadvisor, odepde, parametricsol, particularsol, phaseportrait, poincare, polysols, power\_equivalent, rational\_equivalent, ratsols, redode, reduceOrder, reduce\_order, regular\_parts, regularsp, remove\_RootOf, riccati\_system, riccatisol, rifread, rifsimp, rightdivision, rtaylor, separablesol, singularities, solve\_group, super\_reduce, symgen, symmetric\_power, symmetric\_product, symtest, transinv, translate, untranslate, varparam, zoom]

**> with(VectorCalculus)**

[&x, `\*`, `+`, `-`, `.` , <,>, <|>, About, AddCoordinates, ArcLength, BasisFormat, Binormal, ConvertVector, CrossProduct, Curl, Curvature, D, Del, DirectionalDiff, Divergence, DotProduct, Flux, GetCoordinateParameters, GetCoordinates, GetNames, GetPVDDescription, GetRootPoint, GetSpace, Gradient, Hessian, IsPositionVector, IsRootedVector, IsVectorField, Jacobian, Laplacian, LineInt, MapToBasis, ∇, Norm, Normalize, PathInt, PlotPositionVector, PlotVector, PositionVector, PrincipalNormal, RadiusOfCurvature, RootedVector, ScalarPotential, SetCoordinateParameters, SetCoordinates, SpaceCurve, SurfaceInt, TNBFrame, TangentLine, TangentPlane, TangentVector, Torsion, Vector, VectorField, VectorPotential, VectorSpace, Wronskian, diff, eval, evalVF, int, limit, series] **(3)**

**> JM:=Jacobian([2\*x-x^2-x\*y,-y+x\*y],[x,y]);**

$$JM := \begin{bmatrix} -2x - y + 2 & -x \\ y & x - 1 \end{bmatrix} \quad (4)$$

**> J1:=subs([x=0,y=0],JM);**

$$J1 := \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix} \quad (5)$$

**> # Non-hyperbolic vs HYPERBOLIC equilibrium -> its real part is not zero**

**> # bii)**

```
> eigenvalues(J1)
```

$$2, -1$$

(6)

```
> J1:=subs([x=0,y=0],JM);
```

$$J1 := \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$$

(7)

```
> # there are 3 equilibrium points => 6 cases
```

```
> J2:=subs([x=2,y=1],JM)
```

$$J2 := \begin{bmatrix} -3 & -2 \\ 1 & 1 \end{bmatrix}$$

(8)

```
> eigenvalues(J2)
```

$$\sqrt{2} - 1, -1 - \sqrt{2}$$

(9)

```
> J3:=subs([x=1,y=1],JM)
```

$$J3 := \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}$$

(10)

```
> eigenvalues(J3)# focus is stable attractor
```

$$-\frac{1}{2} + \frac{i\sqrt{3}}{2}, -\frac{1}{2} - \frac{i\sqrt{3}}{2}$$

(11)

```
> # focus is stable attractor
```

```
> restart;
```

```
> eq1:=diff(x(t),t)=2*x(t)
```

$$eq1 := \frac{d}{dt} x(t) = 2x(t)$$

(12)

```
> eq2:=diff(y(t),t)=-y(t)
```

$$eq2 := \frac{d}{dt} y(t) = -y(t)$$

(13)

```
> sist:=eq1,eq2
```

$$sist := \frac{d}{dt} x(t) = 2x(t), \frac{d}{dt} y(t) = -y(t)$$

(14)

```
> cond1:=x(0)=1,y(0)=0
```

$$cond1 := x(0) = 1, y(0) = 0$$

(15)

```
> dsolve({sist,cond1},{x(t),y(t)})
```

$$\{x(t) = e^{2t}, y(t) = 0\}$$

(16)

```
> # the third system:
```

```
> # biv) (0,0) unstable, (2,0) unstable, (1,1) stable
```

```
> # bv)
```

```
> eq11:=diff(x(t),t) = -2*x(t) -2*y(t);
```

$$eq11 := \frac{d}{dt} x(t) = -2x(t) - 2y(t) \quad (17)$$

> eq22:=diff(y(t),t)=y(t)

$$eq22 := \frac{d}{dt} y(t) = y(t) \quad (18)$$

> sist1:=eq11,eq22

$$sist1 := \frac{d}{dt} x(t) = -2x(t) - 2y(t), \frac{d}{dt} y(t) = y(t) \quad (19)$$

> cond2:=x(0)=1,y(0)=0

$$cond2 := x(0) = 1, y(0) = 0 \quad (20)$$

> # numeric vs symbolic vs NEURAL NETWORK solution

> # numeric = a vector of points -> integrate for each sub-interval  
(determined by step size) -> approximate the original function

> dsolve({sist1,cond2},{x(t),y(t)})

$$\{x(t) = e^{-2t}, y(t) = 0\} \quad (21)$$

> eq111:=diff(x(t),t)=-x(t)-y(t)

$$eq111 := \frac{d}{dt} x(t) = -x(t) - y(t) \quad (22)$$

> eq222:=diff(y(t),t)=x(t)

$$eq222 := \frac{d}{dt} y(t) = x(t) \quad (23)$$

> restart

> with(linalg)

[BlockDiagonal, GramSchmidt, JordanBlock, LUdecomp, QRdecomp, Wronskian, addcol, addrow, adj, adjoint, angle, augment, backsub, band, basis, bezout, blockmatrix, charmat, charpoly, cholesky, col, coldim, colspace, colspan, companion, concat, cond, copyinto, crossprod, curl, definite, delcols, delrows, det, diag, diverge, dotprod, eigenvals, eigenvalues, eigenvectors, eigenvects, entermatrix, equal, exponential, extend, ffgausselim, fibonacci, forwardsub, frobenius, gausselim, gaussjord, geneqns, genmatrix, grad, hadamard, hermite, hessian, hilbert, htranspose, ihermite, indexfunc, innerprod, intbasis, inverse, ismith, issimilar, iszero, jacobian, jordan, kernel, laplacian, leastsqrs, linsolve, matadd, matrix, minor, minpoly, mulcol, mulrow, multiply, norm, normalize, nullspace, orthog, permanent, pivot, potential, randmatrix, randvector, rank, ratform, row, rowdim, rowspace, rowspan, rref, scalarmul, singularvals, smith, stackmatrix, submatrix, subvector, sumbasis, swapcol, swaprow, sylvester, toeplitz, trace, transpose, vandermonde, vecpotent, vectdim, vector, wronskian] (24)

**> with(DEtools)**

**(25)**

[AreSimilar, Closure, DENormal, DEplot, DEplot3d, DEplot\_polygon, DFactor, DFactorLCLM, DFactorsols, Dchangevar, Desingularize, FindODE, FunctionDecomposition, GCRD, Gosper, Heunsols, Homomorphisms, IVPsol, IsHyperexponential, LCLM, MeijerGsols, MultiplicativeDecomposition, ODEInvariants, PDEchangecoords, PolynomialNormalForm, RationalCanonicalForm, ReduceHyperexp, RiemannPsols, Xchange, Xcommutator, Xgauge, Zeilberger, abelsol, adjoint, autonomous, bernoullisol, buildsol, buildsym, canoni, caseplot, casesplit, checkrank, chinisol, clairautsol, constcoeffsols, convertAlg, convertsys, dalembertsol, dcoeffs, de2diffop, dfieldplot, diff\_table, diffop2de, dperiodic\_sols, dpolyform, dsubs, eigenring, endomorphism\_charpoly, equinv, eta\_k, eulersols, exactsol, expsols, exterior\_power, firint, firtest, formal\_sol, gen\_exp, generate\_ic, genhomosol, gensys, hamilton\_eqs, hypergeometricsols, hypergeomsols, hyperode, indicialeq, infgen, initialdata, integrate\_sols, intfactor, invariants, kovacicsols, leftdivision, liesol, line\_int, linearsol, matrixDE, matrix\_riccati, maxdimsystems, moser\_reduce, muchange, mult, mutest, newton\_polygon, normalG2, ode\_int\_y, ode\_y1, odeadvisor, odepde, parametricsol, particularsol, phaseportrait, poincare, polysols, power\_equivalent, rational\_equivalent, ratsols, redode, reduceOrder, reduce\_order, regular\_parts, regularsp, remove\_RootOf, riccati\_system, riccatisol, rifread, rifsimp, rightdivision, rtaylor, separablesol, singularities, solve\_group, super\_reduce, symgen, symmetric\_power, symmetric\_product, symtest, transinv, translate, untranslate, varparam, zoom]

**> with(VectorCalculus)**

**(26)**

[&x, `\*`, `+`, `-`, `.` , <,>, <|>, About, AddCoordinates, ArcLength, BasisFormat, Binormal, ConvertVector, CrossProduct, Curl, Curvature, D, Del, DirectionalDiff, Divergence, DotProduct, Flux, GetCoordinateParameters, GetCoordinates, GetNames, GetPVDDescription, GetRootPoint, GetSpace, Gradient, Hessian, IsPositionVector, IsRootedVector, IsVectorField, Jacobian, Laplacian, LineInt, MapToBasis,  $\nabla$ , Norm, Normalize, PathInt, PlotPositionVector, PlotVector, PositionVector, PrincipalNormal, RadiusOfCurvature, RootedVector, ScalarPotential, SetCoordinateParameters, SetCoordinates, SpaceCurve, SurfaceInt, TNBFrame, TangentLine, TangentPlane, TangentVector, Torsion, Vector, VectorField,

*VectorPotential, VectorSpace, Wronskian, diff, eval, evalVF, int, limit, series]*

**> # the nonl-linear system**

**> ec1:=diff(x(t),t)=2\*x(t)-x(t)^2-x(t)\*y(t)**

$$ec1 := \frac{d}{dt} x(t) = 2x(t) - x(t)^2 - x(t)y(t) \quad (27)$$

**> ec2:=diff(y(t),t)=-y(t)+x(t)\*y(t)**

$$ec2 := \frac{d}{dt} y(t) = -y(t) + x(t)y(t) \quad (28)$$

**> sist:=ec1,ec2**

$$sist := \frac{d}{dt} x(t) = 2x(t) - x(t)^2 - x(t)y(t), \frac{d}{dt} y(t) = -y(t) + x(t)y(t) \quad (29)$$

**> condin1:=x(0)=1,y(0)=0**

$$condin1 := x(0) = 1, y(0) = 0 \quad (30)$$

**> dsolve({sist,condin1},{x(t),y(t)})**

$$\left\{ x(t) = \frac{2}{1 + e^{-2t}}, y(t) = 0 \right\} \quad (31)$$

**> condin2:=x(0)=1,y(0)=1**

$$condin2 := x(0) = 1, y(0) = 1 \quad (32)$$

**> dsolve({sist,condin2},{x(t),y(t)})**

$$\{x(t) = 1, y(t) = 1\} \quad (33)$$

**> condin3:=x(0)=0,y(0)=1**

$$condin3 := x(0) = 0, y(0) = 1 \quad (34)$$

**> dsolve({sist,condin3},{x(t),y(t)});**

**> condin4:=x(0)=-1,y(0)=1;**

$$condin4 := x(0) = -1, y(0) = 1 \quad (35)$$

**> dsolve({sist,condin4},{x(t),y(t)});**

**> condin5:=x(0)=-1,y(0)=0**

$$condin5 := x(0) = -1, y(0) = 0 \quad (36)$$

**> dsolve({sist,condin5},{x(t),y(t)});**

$$\left\{ x(t) = \frac{2}{1 - 3e^{-2t}}, y(t) = 0 \right\} \quad (37)$$

**> condin6:=x(0)=-1,y(0)=-1**

$$condin6 := x(0) = -1, y(0) = -1 \quad (38)$$

**> dsolve({sist,condin6},{x(t),y(t)})**

**> condin7:=x(0)=0,y(0)=-1**

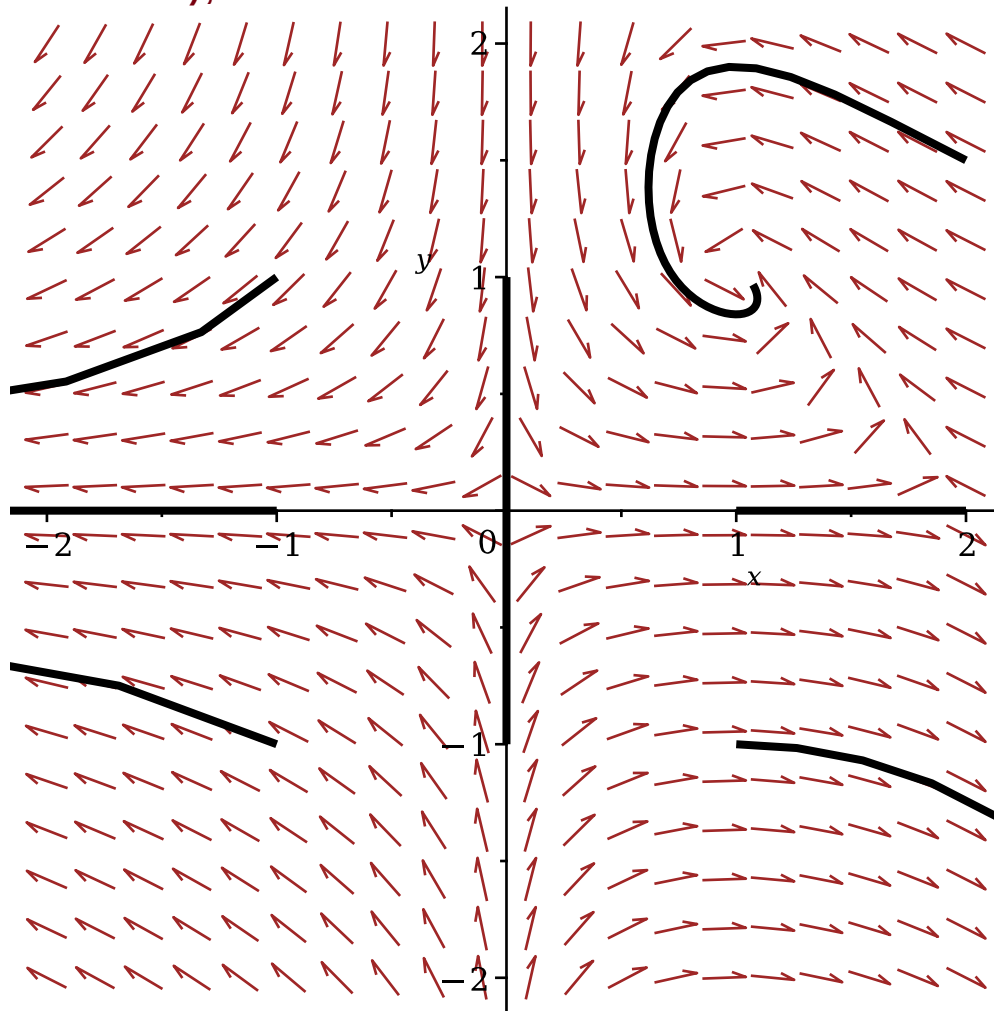
$$condin7 := x(0) = 0, y(0) = -1 \quad (39)$$

**> dsolve({sist,condin7},{x(t),y(t)})**

**> condin8:=x(0)=1,y(0)=-1**

$$condin8 := x(0) = 1, y(0) = -1 \quad (40)$$

```
> dsolve({sist,condin8},{x(t),y(t)})
> DEplot([ec1,ec2],[x(t),y(t)],t=0..6,x=-2..2,y=-2..2,[[x(0)=1,y(0)=0],[x(0)=2,y(0)=1.5],[x(0)=0,y(0)=1],[x(0)=-1,y(0)=1],[x(0)=-1,y(0)=0],[x(0)=-1,y(0)=-1],[x(0)=0,y(0)=-1],[x(0)=1,y(0)=-1]],
linecolor=black);
```



```
> ?DEplot
```

```
> with(DEtools)
```

```
[AreSimilar, Closure, DENormal, DEplot, DEplot3d, DEplot_polygon,
DFactor, DFactorLCLM, DFactorsols, Dchangevar, Desingularize,
FindODE, FunctionDecomposition, GCRD, Gosper, Heunsols,
Homomorphisms, IVPsol, IsHyperexponential, LCLM, MeijerGsols,
MultiplicativeDecomposition, ODEInvariants, PDEchangecoords,
PolynomialNormalForm, RationalCanonicalForm, ReduceHyperexp,
RiemannPsols, Xchange, Xcommutator, Xgauge, Zeilberger, abelsol,
adjoint, autonomous, bernoullisol, buildsol, buildsym, canoni, caseplot,
casesplit, checkrank, chinisol, clairautsol, constcoeffsols, convertAlg,
convertsys, dalembertsol, dcoeffs, de2diffop, dfieldplot, diff_table,
diffop2de, dperiodic_sols, dpolyform, dsubs, eigenring,
```

(41)

*endomorphism\_charpoly, equinv, eta\_k, eulersols, exactsol, expsols,*  
*exterior\_power, firint, firtest, formal\_sol, gen\_exp, generate\_ic,*  
*genhomosol, gensys, hamilton\_eqs, hypergeometricsols, hypergeomsols,*  
*hyperode, indicialeq, infgen, initialdata, integrate\_sols, intfactor,*  
*invariants, kovaciccsols, leftdivision, liesol, line\_int, linearsol, matrixDE,*  
*matrix\_riccati, maxdimsystems, moser\_reduce, muchange, mult, mutest,*  
*newton\_polygon, normalG2, ode\_int\_y, ode\_y1, odeadvisor, odepde,*  
*parametricsol, particularsol, phaseportrait, poincare, polysols,*  
*power\_equivalent, rational\_equivalent, ratsols, redode, reduceOrder,*  
*reduce\_order, regular\_parts, regularsp, remove\_RootOf, riccati\_system,*  
*riccatisol, rifread, rifsimp, righdivision, rtaylor, separablesol,*  
*singularities, solve\_group, super\_reduce, symgen, symmetric\_power,*  
*symmetric\_product, symtest, transinv, translate, untranslate, varparam,*  
*zoom]*

**> dfieldplot([ec1,ec2],[x,y],t=0..1,x=-1..2.2, y=-1..1.2)**

